Part 1: Requirements

1. High-Level Functional Requirements

The Educational Course Management System is a desktop application for educational institutions. It helps manage freshman student registration, course enrollment, grades, GPA calculation, tuition fee payments, and transcript requests.

The system should allow the following:

1. The registrar/admin can register new freshman students.
2. The registrar/admin can create and manage course records.
3. Students can enroll in available courses.
4. Instructors can record and update student grades.
5. The system can calculate GPA automatically based on recorded grades.
6. The finance/admin staff can record tuition fee payments and check payment status.
7. Students can request transcripts.
8. The registrar/admin can review and approve transcript requests.
9. Authorized users can log in and access only the features related to their role.

Functional Use Cases:

Field	Details
Use case Id	F001
Name	Register Student
Created By/Author	Mohammed Awadh
Date Created	1 Apr 2026
Actor	Registrar/Admin
Description/Summary	The registrar/admin registers a new freshman student by entering student information into the system.
Preconditions	The registrar/admin is logged in.
Postconditions	A new student record is stored in the system.
Normal course of events	1. The registrar/admin opens the registration form. 2. The registrar/admin enters the student details. 3. The registrar/admin clicks Save. 4. The system stores the student record.
Exceptions	Required fields are missing. Student ID already exists.
Acceptance Criteria	The student record is created successfully and can be viewed later.

Field	Details
Use case Id	F002
Name	Manage Courses
Created By/Author	Kareem Abdalla
Date Created	1 Apr 2026
Actor	Registrar/Admin
Description/Summary	The registrar/admin adds, updates, or removes course information.
Preconditions	The registrar/admin is logged in.
Postconditions	Course records are updated in the system.
Normal course of events	1. The registrar/admin opens the course management screen. 2. The registrar/admin adds or edits course details. 3. The registrar/admin saves the changes. 4. The system updates the course record.
Exceptions	Course code already exists. Invalid data is entered.
Acceptance Criteria	The course information is saved correctly and appears in the course list.

Field	Details
Use case Id	F003
Name	Enroll in Course
Created By/Author	Kareem Abdalla
Date Created	1 Apr 2026
Actor	Student
Description/Summary	A student enrolls in an available course.
Preconditions	The student is registered, logged in, and the enrollment period is open.
Postconditions	The enrollment record is stored in the system.
Normal course of events	1. The student opens the enrollment screen. 2. The system shows available courses. 3. The student selects a course. 4. The system checks enrollment rules. 5. The system enrolls the student.
Exceptions	Course is full. Student is already enrolled. Student has a restriction such as unpaid fees.
Acceptance Criteria	The student is enrolled only if all enrollment rules are satisfied.

Field	Details
Use case Id	F004
Name	Record Grades
Created By/Author	Mohammed Awadh
Date Created	1 Apr 2026
Actor	Instructor
Description/Summary	The instructor records or updates grades for students in a course.
Preconditions	The instructor is logged in, assigned to the course, and students are enrolled.
Postconditions	Grades are stored or updated in the system.
Normal course of events	1. The instructor opens the grade entry screen. 2. The instructor selects the course. 3. The system shows enrolled students. 4. The instructor enters or updates grades. 5. The instructor saves. 6. The system stores the grades.
Exceptions	Invalid grade value is entered. Instructor is not assigned to the course.
Acceptance Criteria	Grades are saved correctly for the correct students and course.

Field	Details
Use case Id	F005
Name	Calculate GPA
Created By/Author	Mohammed Awadh
Date Created	1 Apr 2026
Actor	System
Description/Summary	The system calculates GPA using student grades and course credit hours.
Preconditions	Student grades are available.
Postconditions	GPA is calculated and stored in the student record.
Normal course of events	1. The system reads grade records and credit hours. 2. The system applies the GPA formula. 3. The system stores the GPA. 4. The GPA is displayed in the student record.
Exceptions	No completed courses are available for calculation.
Acceptance Criteria	The GPA is calculated correctly based on the stored academic data.

Field	Details
Use case Id	F006
Name	Record Tuition Payment
Created By/Author	Kareem Abdalla
Date Created	1 Apr 2026
Actor	Finance Staff
Description/Summary	Finance staff records a tuition payment and updates the payment status.
Preconditions	Finance staff is logged in and the student record exists.
Postconditions	Payment information is stored and the student balance is updated.
Normal course of events	1. Finance staff opens the payment screen. 2. Finance staff selects the student. 3. Finance staff enters payment details. 4. Finance staff clicks Save. 5. The system stores the payment and updates the balance.
Exceptions	Invalid amount is entered. Student record is not found.
Acceptance Criteria	The payment is saved correctly and the updated balance is shown.

Field	Details
Use case Id	F007
Name	Request Transcript
Created By/Author	Kareem Abdalla
Date Created	1 Apr 2026
Actor	Student
Description/Summary	A student submits a request for an academic transcript.
Preconditions	The student is logged in and the student record exists.
Postconditions	A transcript request is stored with pending status.
Normal course of events	1. The student opens the transcript request screen. 2. The student enters the required request details. 3. The student clicks Submit. 4. The system stores the request.
Exceptions	Required request information is missing. Transcript release is blocked due to unpaid fees.
Acceptance Criteria	The transcript request is stored successfully for admin review.

Field	Details
Use case Id	F008
Name	Approve Transcript Request
Created By/Author	Mohammed Awadh
Date Created	1 Apr 2026
Actor	Registrar/Admin
Description/Summary	The registrar/admin reviews a transcript request and approves or rejects it.
Preconditions	The registrar/admin is logged in and a transcript request exists.
Postconditions	The request status is updated in the system.
Normal course of events	1. The registrar/admin opens the transcript request list. 2. The registrar/admin selects a request. 3. The system shows the request details. 4. The registrar/admin approves or rejects the request. 5. The system updates the request status.
Exceptions	Request not found. Student is not eligible for transcript release.
Acceptance Criteria	The transcript request status is saved correctly as approved or rejected.

Field	Details
Use case Id	F009
Name	User Login
Created By/Author	Kareem Abdalla
Date Created	1 Apr 2026
Actor	Student / Registrar/Admin / Instructor / Finance Staff
Description/Summary	A user logs into the system using valid credentials.
Preconditions	A user account exists.
Postconditions	The user gains access according to their role.
Normal course of events	1. The user enters username and password. 2. The user clicks Login. 3. The system checks the credentials. 4. The system opens the correct dashboard.
Exceptions	Invalid username or password. Account is locked or inactive.
Acceptance Criteria	Only valid users can access the system and only role-allowed features are shown.

Abuse cases:

Field	Details
Abuse case Id	A001
Name	Unauthorized Login
Created By/Author	Kareem
Date	1 Apr 2026

Field	Details
Priority	High
Scope	Educational Course Management System
Development Environment	Desktop application
Mis-actors	Malicious user
Access Right Levels	Other Network User / Low Level System Users
Point of Entry	Login screen
Security Attributes Affected	Confidentiality, Integrity
Description	A malicious user tries to enter the system using guessed or stolen credentials.
Sophistication	Simple
Preconditions	The login screen is accessible.
Assumptions	The attacker knows or guesses a valid username.
Postconditions	The attacker gains unauthorized access to the system.
Related Use Cases	F009 / User Login
Related Threats	Unauthorized access, password guessing
Exceptions	N/A
Acceptance Criteria	The attacker successfully logs into the system without valid authorization.

Field	Details
Abuse case Id	A002
Name	Invalid Course Enrollment
Created By/Author	Mohammed Awadh
Date	1 Apr 2026
Priority	Medium
Scope	Course enrollment module
Development Environment	Desktop application
Mis-actors	Malicious student
Access Right Levels	Low Level System Users
Point of Entry	Course enrollment screen
Security Attributes Affected	Integrity
Description	A student tries to enroll in a course without meeting the required rules.
Sophistication	Simple
Preconditions	The student is logged in.
Assumptions	The student can access the enrollment screen.
Postconditions	The student is incorrectly enrolled in the course.
Related Use Cases	F003 / Enroll in Course
Related Threats	Business rule bypass
Exceptions	N/A

Field	Details
Acceptance Criteria	The student is added to a course even though the course rules are not met.

Field	Details
Abuse case Id	A003
Name	Unauthorized Grade Change
Created By/Author	Mohammed Awadh
Date	1 Apr 2026
Priority	High
Scope	Grade management module
Development Environment	Desktop application
Mis-actors	Unauthorized staff user
Access Right Levels	Low/Medium Level System Users
Point of Entry	Grade entry screen
Security Attributes Affected	Integrity, Confidentiality
Description	A user who is not the assigned instructor changes student grades.
Sophistication	Medium
Preconditions	The user is logged in.
Assumptions	The user can reach the grade entry screen.
Postconditions	Student grade records are changed without permission.

Field	Details
Related Use Cases	F004 / Record Grades
Related Threats	Unauthorized modification of data
Exceptions	N/A
Acceptance Criteria	Grades are modified by an unauthorized user.

Field	Details
Abuse case Id	A004
Name	Fake Payment Entry
Created By/Author	Mohammed Awadh
Date	1 Apr 2026
Priority	High
Scope	Tuition payment module
Development Environment	Desktop application
Mis-actors	Unauthorized staff user
Access Right Levels	Medium Level System Users / System Admin
Point of Entry	Payment entry form
Security Attributes Affected	Integrity
Description	A user enters false payment information to make it appear that tuition was paid.
Sophistication	Medium

Field	Details
Preconditions	The user can access the payment module.
Assumptions	Payment details are entered manually.
Postconditions	Incorrect payment information is saved.
Related Use Cases	F006 / Record Tuition Payment
Related Threats	Data tampering, fraud
Exceptions	N/A
Acceptance Criteria	A false payment record is stored successfully.

Field	Details
Abuse case Id	A005
Name	Unauthorized Transcript Access
Created By/Author	Kareem Abdalla
Date	1 Apr 2026
Priority	High
Scope	Transcript request and release module
Development Environment	Desktop application
Mis-actors	Malicious user
Access Right Levels	Other Network User / Low Level System Users
Point of Entry	Transcript request or transcript viewing screen
Security Attributes Affected	Confidentiality
Description	A user tries to view or obtain another student's transcript without permission.
Sophistication	Medium
Preconditions	Transcript records exist in the system.
Assumptions	The attacker can access the transcript section.
Postconditions	Transcript data is exposed to an unauthorized person.
Related Use Cases	F007 / Request Transcript, F008 / Approve Transcript Request

Field	Details
Related Threats	Privacy breach, unauthorized data access
Exceptions	N/A
Acceptance Criteria	An unauthorized user is able to access transcript information.

Security Use Cases:

Field	Details
Use case Id	S001
Name	Authenticate User Login
Created By/Author	Kareem Abdalla
Date Created	1 Apr 2026
Actor	System
Description/Summary	The system must verify login credentials before allowing any user to access the system.
Preconditions	A user attempts to log in.
Postconditions	Only authenticated users are allowed into the system.
Normal course of events	1. The user enters username and password. 2. The system checks the credentials. 3. The system grants access if valid. 4. The system denies access if invalid.
Exceptions	None
Acceptance Criteria	Invalid login attempts are rejected and valid users are granted access.

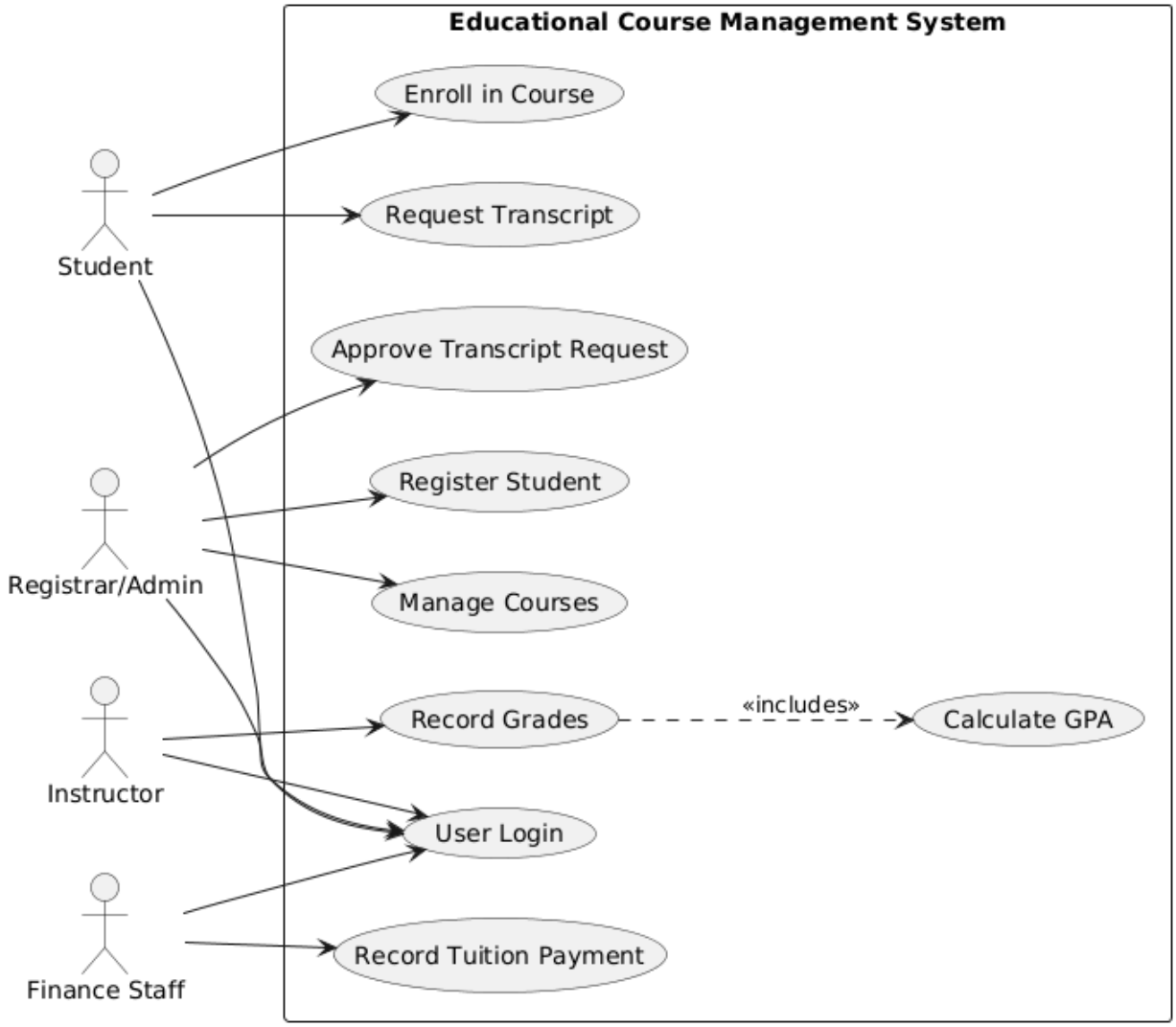
Field	Details
Use case Id	S002
Name	Validate Enrollment Rules
Created By/Author	Kareem Abdalla
Date Created	1 Apr 2026
Actor	System
Description/Summary	The system must check enrollment rules before confirming student enrollment.
Preconditions	A student selects a course.
Postconditions	The student is enrolled only if all rules are satisfied.
Normal course of events	1. The student selects a course. 2. The system checks whether the course is full. 3. The system checks whether the student is already enrolled. 4. The system checks enrollment restrictions. 5. The system accepts or rejects the request.
Exceptions	None
Acceptance Criteria	The system blocks enrollment when the student does not meet the course requirements.

Field	Details
Use case Id	S003
Name	Restrict Grade Update Access
Created By/Author	Mohammed Awadh
Date Created	1 Apr 2026
Actor	System
Description/Summary	The system must allow grade entry and updates only for authorized users.
Preconditions	A user tries to open or use the grade entry function.
Postconditions	Unauthorized users cannot view or modify grade data.
Normal course of events	1. A user opens the grade entry screen. 2. The system checks the user role and assignment. 3. The system allows access only if the user is authorized. 4. The system blocks unauthorized changes.
Exceptions	None
Acceptance Criteria	Only authorized instructors or approved admin users can update grades.

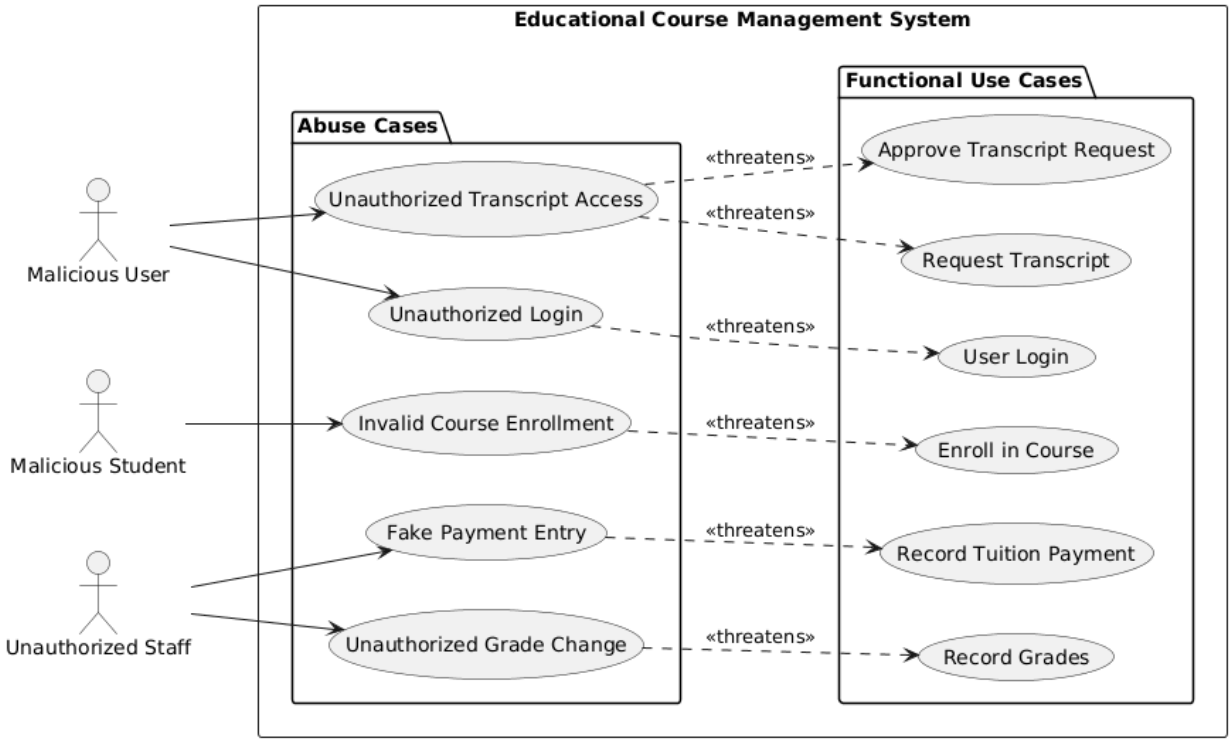
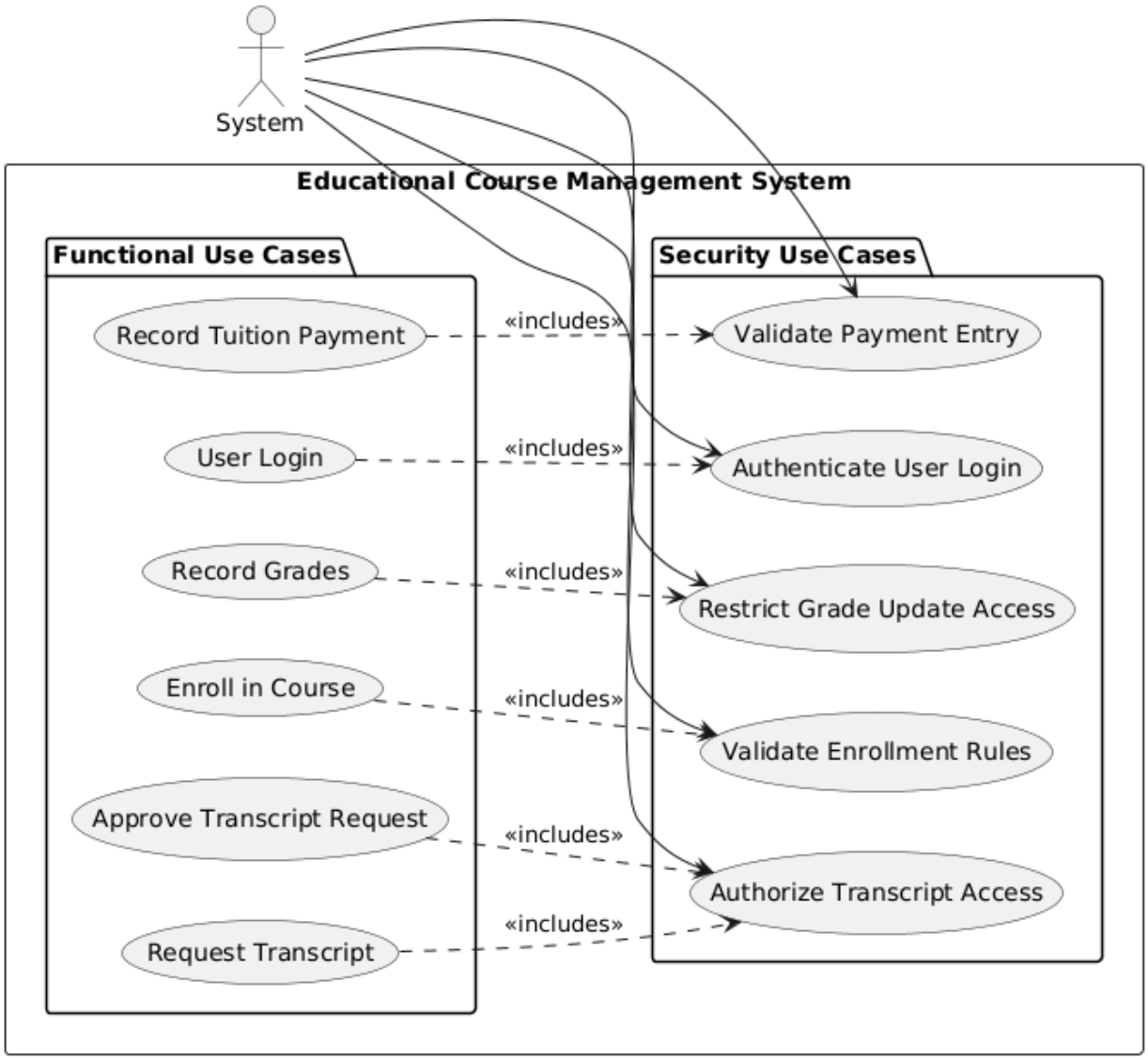
Field	Details
Use case Id	S004
Name	Validate Payment Entry
Created By/Author	Mohammed Awadh
Date Created	1 Apr 2026
Actor	System
Description/Summary	The system must validate payment details before storing tuition payment records.
Preconditions	Finance staff enters payment information.
Postconditions	Only valid payment records are stored.
Normal course of events	1. Finance staff enters payment details. 2. The system checks required fields. 3. The system checks that the amount is valid. 4. The system saves the payment if valid. 5. The system rejects invalid entries.
Exceptions	None
Acceptance Criteria	The system prevents incomplete or invalid payment records from being stored.

Field	Details
Use case Id	S005
Name	Authorize Transcript Access
Created By/Author	Mohammed Awadh
Date Created	1 Apr 2026
Actor	System
Description/Summary	The system must make sure transcript records are viewed or released only by authorized users.
Preconditions	A user tries to access transcript information.
Postconditions	Transcript data is protected from unauthorized access.
Normal course of events	1. A user opens the transcript section. 2. The system checks user identity and role. 3. The system checks whether the transcript belongs to the student or an authorized admin is handling it. 4. The system allows or denies access.
Exceptions	None
Acceptance Criteria	Unauthorized users cannot view, approve, or release transcript data.

Functional Use Case Diagram:



Abuse Case Diagram:



Security Case Diagram: